

## COST SHEET

### 1. Cost Sheet

It is a statement which shows the break-up and build-up of costs for a particular period (functional classification).

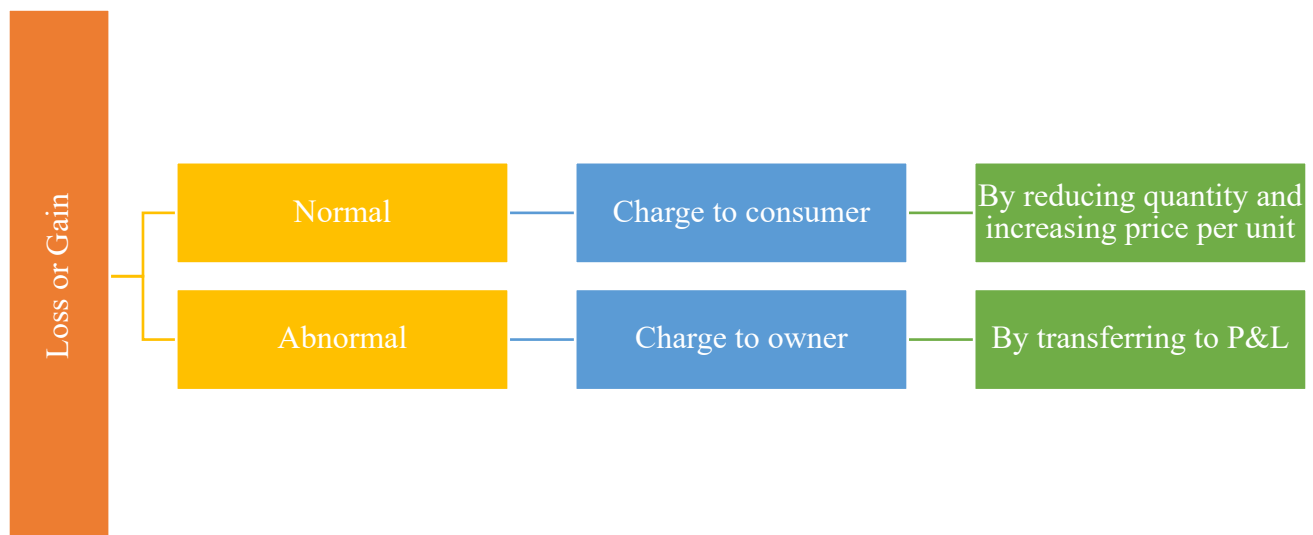
### 2. Statement of Cost/Cost Sheet

| Particulars                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Total Cost<br>(₹) | Cost per<br>unit (₹) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------|
| Opening stock of Raw Material                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                   |                      |
| Add: Purchases                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                      |
| Less: Closing stock of Raw Material                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                   |                      |
| Add: Carriage/Freight inward                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |                      |
| Less: Raw material purchase return                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                   |                      |
| Less: Sale value of scrap of raw material                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                   |                      |
| <b>Direct Material Consumed</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |                      |
| Add: Direct labour cost <ul style="list-style-type: none"> <li>➤ Wages</li> <li>➤ Bonus</li> <li>➤ Allowances</li> <li>➤ Overtime</li> <li>➤ Employer contribution to PF/ESI/SSS</li> <li>➤ Any other benefit</li> </ul>                                                                                                                                                                                                                                                                                                                                                                 |                   |                      |
| Add: Direct expenses or chargeable expenses <ul style="list-style-type: none"> <li>➤ Royalty for production</li> <li>➤ Cost of utilities such as power &amp; fuel, steam etc.</li> <li>➤ Fee for technical know-how</li> <li>➤ Cost of product/service specific design or drawing</li> <li>➤ Cost of product/service specific software</li> <li>➤ Amortized cost of moulds, patterns, patents etc.</li> <li>➤ Job charges paid to job workers</li> <li>➤ Hire charges paid for hiring specific equipment</li> <li>➤ Other expenses which are directly related with production</li> </ul> |                   |                      |
| <b>Prime Cost</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                   |                      |
| Add: Factory/Work Overheads                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |                      |
| <b>Gross Factory/Work Cost</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                   |                      |
| Add: Opening stock of WIP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                   |                      |
| Less: Closing stock of WIP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                   |                      |
| <b>Net Factory/Work cost</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |                      |

| Particulars                                             | Total Cost<br>(₹) | Cost per<br>unit (₹) |
|---------------------------------------------------------|-------------------|----------------------|
| Add: Quality control cost                               |                   |                      |
| Add: Research and Development cost                      |                   |                      |
| Add: Administrative overheads (related to production)   |                   |                      |
| Add: Packing cost (primary)                             |                   |                      |
| Less: Credit for recoveries/Scrap/Defectives/By-Product |                   |                      |
| <b>Cost of Production</b>                               |                   |                      |
| Add: Opening stock of finished goods                    |                   |                      |
| Less: Closing stock of finished goods                   |                   |                      |
| <b>Cost of Goods Sold</b>                               |                   |                      |
| Add: Administrative overheads (general)                 |                   |                      |
| Add: Selling and distribution overheads                 |                   |                      |
| Add: Interest on borrowed funds                         |                   |                      |
| <b>Cost of Sales</b>                                    |                   |                      |
| Add: Profit                                             |                   |                      |
| <b>Sales</b>                                            |                   |                      |

### 3. Points to Remember (PTR)

#### (A) Loss or Gain



#### (B) Certain expenses not appear in cost sheet:

- Goodwill or preliminary expenses written off
- Income tax
- Loss on sale of assets or investment
- Cost pertaining to or arising out of a pandemic e.g. COVID-19
- Penalty, fines, damages etc.

**(C) Work = Factory**

Work Overheads = Factory Overheads

Work Cost = Net Factory Cost

**(D) Cost of goods available for sale** = Opening stock of FG + Cost of Production (COP)

**4. Treatment of Expenses**

| Expenses                                                                   | Treatment                           |
|----------------------------------------------------------------------------|-------------------------------------|
| Insurance on stock of Raw material & WIP                                   | Factory OHs                         |
| Stores Related Expenses<br>Stores Consumed = Opening + Purchases – Closing | Factory OHs                         |
| Warehouse or Godown Expenses                                               | Selling & Distribution OHs          |
| Loose tools written off                                                    | Factory OHs                         |
| Bank charges                                                               | Administration OHs                  |
| Salesmen commission                                                        | Selling & Distribution OHs          |
| Cost of Samples                                                            | Selling & Distribution OHs          |
| Audit Fee                                                                  | Administration OHs                  |
| General Expenses                                                           | Administration OHs                  |
| Counting House Salaries                                                    | Administration OHs                  |
| Production planning expenses in office                                     | Administration OHs related to Prod. |
| Director's fees                                                            | Administration OHs                  |
| Fee for exhibition participation                                           | Selling & Distribution OHs          |
| Pollution control expenses                                                 | Factory OHs                         |
| Carriage on raw material return                                            | Factory OHs                         |
| Bad Debts                                                                  | Ignore                              |
| Packaging                                                                  |                                     |
| (A) Primary Packaging                                                      | Add after NFC                       |
| (B) Secondary Packaging                                                    | Selling & Distribution OHs          |
|                                                                            |                                     |
| GST                                                                        |                                     |
| (A) GST Output                                                             | Add after sales value               |

|                                                         |                                             |
|---------------------------------------------------------|---------------------------------------------|
| (B) GST Input                                           |                                             |
| (C) ITC Available                                       | Ignore                                      |
| (D) ITC Not Available                                   | Add to the cost of item to which it relates |
|                                                         |                                             |
| Custom Duty                                             | Add to the cost of item to which it relates |
|                                                         |                                             |
| Discount                                                |                                             |
| (A) Trade Discount                                      | Deduct if not already deducted              |
| (B) Cash Discount                                       | Ignore                                      |
| (C) Other discount or discount on sales                 | Selling & Distribution OHs                  |
|                                                         |                                             |
| Waste/Scrap                                             |                                             |
| (A) Scrap for which amount is received on sale          |                                             |
| (1) Related to raw material                             | Deduct from raw material consumed           |
| (2) Arises during production                            | Deduct after NFC                            |
| (B) Scrap for which disposal cost is to be incurred     | Add to factory OHs                          |
|                                                         |                                             |
|                                                         |                                             |
| Defectives                                              |                                             |
| (a) Sold as it is at discount                           | Deduct after NFC                            |
| (b) Goods are rectified by incurring rectification cost | Add rectification cost to Factory OHs       |

### 5. Administration Overheads

If administration overheads is \_\_\_% of NFC or If administration overheads is ₹ \_\_\_ per unit produce then in both situation consider them as related to production.

### 6. Conversion Cost

It is the cost to convert raw material into finished goods. It is sum total of direct labour, direct expenses and factory overheads.

**Question – 1**

SK Ltd. has the following expenditures for the year ended 31<sup>st</sup> March:

| Particulars                                                                                  | Amount (₹) | Amount (₹)   |
|----------------------------------------------------------------------------------------------|------------|--------------|
| Raw materials purchased                                                                      |            | 10,00,00,000 |
| GST paid on the above purchases @18% (eligible for input tax credit)                         |            | 1,80,00,000  |
| Freight inwards                                                                              |            | 11,20,600    |
| Wages paid to factory workers                                                                |            | 29,20,000    |
| Contribution made towards employees' PF and ESI                                              |            | 3,60,000     |
| Production bonus paid to factory workers                                                     |            | 2,90,000     |
| Royalty paid for production                                                                  |            | 1,72,600     |
| Amount paid for power & fuel                                                                 |            | 4,62,000     |
| Amount paid for purchase of moulds and patterns (life is equivalent to two years production) |            | 8,96,000     |
| Job charges paid to job workers                                                              |            | 8,12,000     |
| Stores and spares consumed                                                                   |            | 1,12,000     |
| Depreciation on:                                                                             |            |              |
| Factory building                                                                             | 84,000     |              |
| Office building                                                                              | 56,000     |              |
| Plant & Machinery                                                                            | 1,26,000   |              |
| Delivery vehicles                                                                            | 86,000     | 3,52,000     |
| Salary paid to supervisors                                                                   |            | 1,26,000     |
| Repairs & maintenance paid for:                                                              |            |              |
| Plant & Machinery                                                                            | 48,000     |              |
| Sales office building                                                                        | 18,000     |              |
| Vehicles used by directors                                                                   | 19,600     | 85,600       |
| Insurance premium paid for:                                                                  |            |              |
| Plant & Machinery                                                                            | 31,200     |              |
| Factory building                                                                             | 18,100     |              |
| Stock of raw materials & WIP                                                                 | 36,000     | 85,300       |
| Expenses paid for quality control check activities                                           |            | 19,600       |
| Salary paid to quality control staffs                                                        |            | 96,200       |
| Research & development cost paid for improvement in production process                       |            | 18,200       |
| Expenses paid for pollution control and engineering & maintenance                            |            | 26,600       |
| Expenses paid for administration of factory work                                             |            | 1,18,600     |
| Salary paid to functional managers:                                                          |            |              |
| Production control                                                                           | 9,60,000   |              |

|                                                             |           |           |
|-------------------------------------------------------------|-----------|-----------|
| Finance & accounts                                          | 9,18,000  |           |
| Sales & Marketing                                           | 10,12,000 | 28,90,000 |
| Salary paid to General Manager                              |           | 12,56,000 |
| Packaging cost paid for:                                    |           |           |
| Primary packing necessary to maintain quality               | 96,000    |           |
| For re-distribution of finished goods                       | 1,12,000  | 2,08,000  |
| Wages of employees engaged in distribution of goods         |           | 7,20,000  |
| Fee paid to auditors                                        |           | 1,80,000  |
| Fee paid to legal advisors                                  |           | 1,20,000  |
| Fee paid to independent directors                           |           | 2,20,000  |
| Performance bonus paid to sales staff                       |           | 1,80,000  |
| Value of stock as on 1 <sup>st</sup> April of last year     |           |           |
| Raw materials                                               | 18,00,000 |           |
| Work-in-process                                             | 9,20,000  |           |
| Finished goods                                              | 11,00,000 | 38,20,000 |
| Value of stock as on 31 <sup>st</sup> March of current year |           |           |
| Raw materials                                               | 9,60,000  |           |
| Work-in-process                                             | 8,70,000  |           |
| Finished goods                                              | 18,00,000 | 36,30,000 |

Amount realized by selling of scrap and waste generated during manufacturing process is ₹ 86,000.  
From the above data you are required to prepare statement of cost for the year ended 31<sup>st</sup> March, showing (i) prime cost, (ii) factory cost, (iii) cost of production, (iv) cost of goods sold and (v) cost of sales.

**Solution**

| Particulars                              |                 | Amount (₹)   |
|------------------------------------------|-----------------|--------------|
| Opening stock of raw material            |                 | 18,00,000    |
| Add: Raw material purchases              |                 | 10,00,00,000 |
| Less: Closing stock of raw material      |                 | (9,60,000)   |
| Add: Freight inwards                     |                 | 11,20,600    |
| Raw material consumed                    |                 | 10,19,60,600 |
| Direct Labour:                           |                 |              |
| Wages paid to factory workers            | 29,20,000       |              |
| Contribution to PF & ESI                 | 3,60,000        |              |
| Production bonus paid to factory workers | <u>2,90,000</u> | 35,70,000    |
| Direct Expenses:                         |                 |              |
| Royalty paid for production              | 1,72,600        |              |
| Amount paid for power & fuel             | 4,62,500        |              |
| Amortised cost of moulds and patterns    | 4,48,000        |              |

|                                                       |                 |              |
|-------------------------------------------------------|-----------------|--------------|
| Job charges paid to job workers                       | <u>8,12,000</u> | 35,70,000    |
| Prime Cost                                            |                 | 10,74,25,200 |
| Factory overheads:                                    |                 |              |
| Stores and spares consumed                            | 1,12,000        |              |
| Depreciation on factory building                      | 84,000          |              |
| Depreciation on plant & machinery                     | 1,26,000        |              |
| Repairs & maintenance for plant & machinery           | 48,000          |              |
| Insurance premium paid for plant & machinery          | 31,200          |              |
| Insurance premium paid for factory building           | 18,100          |              |
| Insurance premium paid for stock of raw material      | 36,000          |              |
| Salary paid to supervisors                            | 1,26,000        |              |
| Expenses paid for pollution control                   | <u>26,600</u>   | 6,07,900     |
| Gross Factory cost                                    |                 | 10,83,33,100 |
| Add: Opening WIP                                      |                 | 9,20,000     |
| Less: Closing WIP                                     |                 | (8,70,000)   |
| Net Factory cost                                      |                 | 10,80,83,100 |
| Quality control cost:                                 |                 |              |
| Expenses paid for quality control check               | 19,600          |              |
| Salary paid to quality control staff                  | <u>96,200</u>   | 1,15,800     |
| Research and development cost paid                    |                 | 18,200       |
| Administrative overheads related to production        |                 |              |
| Expenses paid for administration                      | 1,18,600        |              |
| Salary paid to production control manager             | <u>9,60,000</u> | 10,78,600    |
| Less: Realisable value on sale of scrap               |                 | (86,000)     |
| Add: Primary packaging cost                           |                 | 96,000       |
| Cost of production                                    |                 | 10,93,05,700 |
| Add: Opening stock of finished goods                  |                 | 11,00,000    |
| Less: Closing stock of finished goods                 |                 | (18,00,000)  |
| Cost of goods sold                                    |                 | 10,86,05,700 |
| Administrative overheads:                             |                 |              |
| Depreciation on office building                       | 56,000          |              |
| Repairs & maintenance paid for vehicles for directors | 19,600          |              |
| Salary paid to manager-finance and accounts           | 9,18,000        |              |
| Salary paid to general manager                        | 12,56,000       |              |
| Fee paid to auditors                                  | 1,80,000        |              |
| Fee paid to legal advisors                            | 1,20,000        |              |
| Fee paid to independent directors                     | <u>2,20,000</u> | 27,69,600    |
| Selling and distribution overheads                    |                 |              |

|                                                     |                 |              |
|-----------------------------------------------------|-----------------|--------------|
| Repairs & maintenance paid to sales office building | 18,000          |              |
| Salary paid to manager – sales & marketing          | 10,12,000       |              |
| Performance bonus paid to sales staffs              | 1,80,000        |              |
| Depreciation on delivery vehicles                   | 86,000          |              |
| Packaging cost paid for re-distribution             | 1,12,000        |              |
| Wages of employees engaged in distribution of goods | <u>7,20,000</u> | 21,28,000    |
| Cost of Sales                                       |                 | 11,35,03,300 |

**Question – 2**

The following data relates to manufacturing of a standard product during the month of the March:

| Particulars                                         | Amount (in ₹) |
|-----------------------------------------------------|---------------|
| Stock of Raw material as on 01-03                   | 80,000        |
| Work in progress as on 01-03                        | 50,000        |
| Purchase of raw material                            | 2,00,000      |
| Carriage inwards                                    | 20,000        |
| Direct wages                                        | 1,20,000      |
| Cost of special drawing                             | 30,000        |
| Hire charges paid for Plant                         | 24,000        |
| Return of Raw Material                              | 40,000        |
| Carriage on return                                  | 6,000         |
| Expenses for participation in Industrial exhibition | 8,000         |
| Legal charges                                       | 2,500         |
| Salary to office staff                              | 25,000        |
| Maintenance of office building                      | 2,000         |
| Depreciation on Delivery Van                        | 6,000         |
| Warehousing charges                                 | 1,500         |
| Stock of Raw material as on 31-03                   | 30,000        |
| Stock of Work in Progress as on 31-03               | 24,000        |

- Store overheads on material are 10% of material consumed.
- Factory overheads are 20% of the prime cost
- 10% of the output was rejected and a sum of ₹ 5,000 was realized on sale of scrap.
- 10% of the finished product was found to be defective and the defective products were rectified at an additional expenditure which is equivalent to 20% of proportionate direct wages.
- The total output was 8,000 units during the month.

You are required to prepare a cost sheet for the above period showing the:

- Cost of raw material consumed
- Prime cost



- (iii) Work cost
- (iv) Cost of production
- (v) Cost of sales

**Solution****Cost Sheet**

| Particulars                                                             | Amount (₹) |
|-------------------------------------------------------------------------|------------|
| Opening stock of raw material                                           | 80,000     |
| Add: Raw material purchases                                             | 2,00,000   |
| Add: Carriage inward                                                    | 20,000     |
| Less: Return of raw material                                            | (40,000)   |
| Less: Closing stock of raw material                                     | (30,000)   |
| <b>Raw Material consumed</b>                                            | 2,30,000   |
| Direct wages                                                            | 1,20,000   |
| Direct Expenses:    Cost of special drawing                      30,000 |            |
| Hire charges paid for plant <u>24,000</u>                               | 54,000     |
| <b>Prime Cost</b>                                                       | 4,04,000   |
| Stores Overheads (10% × 2,30,000)                                       | 23,000     |
| Carriage on return of raw material                                      | 6,000      |
| Factory overheads (20% × 4,04,000)                                      | 80,800     |
| Rectification cost of defectives (1,20,000 × 90% × 10% × 20%)           | 2,160      |
| <b>Gross Factory Cost</b>                                               | 5,15,960   |
| Add: Opening WIP                                                        | 50,000     |
| Less: Closing WIP                                                       | (24,000)   |
| <b>Net Factory Cost</b>                                                 | 5,41,960   |
| Less: Scrap sale                                                        | (5,000)    |
| <b>Cost of Production/COGS</b>                                          | 5,36,960   |
| Administration Overheads:                                               |            |
| Legal charges                                              2,500        |            |
| Salary to office staff                                      25,000      |            |
| Maintenance of office building <u>2,000</u>                             | 29,500     |
| Selling & Distribution Overheads:                                       |            |
| Expenses for participation in industrial exhibition 8,000               |            |
| Warehousing charges                                      1,500          |            |
| Depreciation on Delivery Van <u>6,000</u>                               | 15,500     |
| <b>Cost of Sales</b>                                                    | 5,81,960   |

**Question – 3**

A fire occurred in the factory premises on October 31. The accounting records have been destroyed.

Certain accounting records were kept in another building. They reveal the following for the period September 1 to October 31:

|                                               |                        |
|-----------------------------------------------|------------------------|
| (a) Direct materials purchased                | ₹ 2,50,000             |
| (b) Work in progress inventory (1 Sep)        | ₹ 40,000               |
| (c) Direct material inventory (1 Sep)         | ₹ 20,000               |
| (d) Finished goods inventory (1 Sep)          | ₹ 37,750               |
| (e) Indirect manufacturing costs              | 40% of conversion cost |
| (f) Sales revenue                             | ₹ 7,50,000             |
| (g) Direct Manufacturing labour               | ₹ 2,22,250             |
| (h) Prime costs                               | ₹ 3,97,750             |
| (i) Gross margin percentage based on revenues | 30%                    |
| (j) Cost of goods available for sale          | ₹ 5,55,775             |

The cost is fully covered by insurance. The insurance company wants to know the historical cost of the inventories as the basis for negotiating a settlement, although the settlement is actually to be based on replacement cost, not historical cost.

Required:

- Finished goods inventory 31 October
- Work in process inventory 31 October
- Direct material inventory 31 October

### Solution

#### Statement of cost and sales

| Particulars                                     | Amount          |
|-------------------------------------------------|-----------------|
| Opening stock of material                       | 20,000          |
| Add: Purchases                                  | 2,50,000        |
| Less: Closing stock of material(bal. fig.)      | (94,500)        |
| <b>Direct material consumed (bal. fig.)</b>     | <b>1,75,500</b> |
| Add: Direct Labour                              | 2,22,250        |
| Add: Direct Expenses                            | -               |
| <b>Prime Cost (given)</b>                       | <b>3,97,750</b> |
| Add: Factory Overheads (working note-1)         | 1,48,167        |
| <b>Gross Factory Cost</b>                       | <b>5,45,917</b> |
| Add: Opening WIP                                | 40,000          |
| <b>Goods Processed during the period</b>        | <b>5,85,917</b> |
| Less: Closing WIP(bal. fig.)                    | (67,892)        |
| <b>Net Factory Cost/COP (bal. fig.)</b>         | <b>5,18,025</b> |
| Add: Opening stock of FG                        | 37,750          |
| <b>Cost of goods available for sale (given)</b> | <b>5,55,775</b> |
| Less: Closing stock of FG(bal. fig.)            | (30,775)        |
| <b>Cost of goods sold</b>                       | <b>5,25,000</b> |
| Add: Administration Overheads                   | -               |
| Add: Selling & Distribution Overheads           | -               |

| Particulars                  | Amount   |
|------------------------------|----------|
| Cost of Sales (Bal. fig.)    | 5,25,000 |
| Add: Profit (7,50,000 × 30%) | 2,25,000 |
| Sales                        | 7,50,000 |

**Working note - 1**

Let factory overheads =  $x$

Thus,  $x = 40\% (2,22,250 + 0 + x)$

$x = 1,48,167$

**7. Valuation of Stock**

Stock can be valued either on FIFO basis or LIFO basis or Weighted average method. Unless otherwise provided FIFO method will be used for valuation of stock.

According to FIFO Method,

Value of cl. stock of raw material =  $\frac{\text{Amount of raw material purchased}}{\text{Raw material purchase quantity}} \times \text{Cl. stock raw material units}$

Value of closing stock of finished goods =  $\frac{\text{Cost of Production}}{\text{Units produced}} \times \text{Closing stock finished goods units}$

According to Weighted Average Method,

Value of cl. stock of raw material =  $\frac{\text{RM purchase} + \text{Op. Stock of RM}}{\text{RM purchase units} + \text{Op. stock RM units}} \times \text{Cl. stock raw mat. units}$

Value of closing stock of finished goods =  $\frac{\text{Cost of Production} + \text{Op. stock FG}}{\text{Units produced} + \text{Op. stock FG units}} \times \text{Closing stock FG units}$

For raw material,

Raw material consumed units = Op. stock RM units + RM Purchase units – Cl. stock RM

For finished goods,

Finished goods units sold = Op. stock FG units + FG units produced – Cl. stock FG units

If there is NIL Opening stock

Cost per unit of finished goods =  $\frac{\text{Cost of Production}}{\text{Units produced}} = \frac{\text{Cost of Goods Sold}}{\text{Units Sold}}$

**8. Calculation of per unit data and vice versa**

(A) For calculating per unit data

- Divide all values by Units Produced upto cost of production
- Divide all values by Units Sold from COGS and onwards

(B) If factory overheads or administration overheads (production related) per unit is given then multiply it with number of units produced to get total value.

(C) If administration overheads (general) and selling and distribution overheads per unit is given then multiply it with number of units sold to get total value.

**Question – 4**

From the following particulars, you are required to prepare monthly cost sheet of SK Ltd.:

|                                                     | Amount (₹)  |
|-----------------------------------------------------|-------------|
| Opening Inventories                                 |             |
| - Raw materials                                     | 12,00,000   |
| - Work-in-process                                   | 18,00,000   |
| - Finished goods (10,000 units)                     | 9,60,000    |
| Closing inventories                                 |             |
| - Raw materials                                     | 14,00,000   |
| - Work-in-process                                   | 16,04,000   |
| - Finished goods                                    | ?           |
| Raw material purchased                              | 1,44,00,000 |
| GST paid on raw materials purchased (ITC available) | 7,20,000    |
| Wages paid to production workers                    | 36,64,000   |
| Expenses paid for utilities                         | 1,45,600    |
| Office and administration expenses paid             | 26,52,000   |
| Travelling allowance paid to office staffs          | 1,21,000    |
| Selling expenses                                    | 6,46,000    |

Machine hours worked – 21,600 hours

Machine hour rate - ₹ 8.00 per hour

Units sold – 1,60,000

Units produced 1,94,000

Desired profit – 15% on sales

**Solution**

**Cost sheet of SK Ltd. for month of .....**

**Units produced – 1,94,000**

**Units sold – 1,60,000**

| Particulars                           | (₹)                | Cost per unit (₹) |
|---------------------------------------|--------------------|-------------------|
| Raw materials purchased               | 1,44,00,000        |                   |
| Add: Opening value of raw materials   | 12,00,000          |                   |
| Less: Closing value of raw materials  | (14,00,000)        |                   |
| <b>Materials consumed</b>             | <b>1,42,00,000</b> | <b>73.19</b>      |
| Wages paid to production workers      | 36,64,000          | 18.89             |
| Expenses paid for utilities           | 1,45,600           | 0.75              |
| <b>Prime Cost</b>                     | <b>1,80,09,600</b> | <b>92.83</b>      |
| Factory overheads (₹ 8 × 2,600 hours) | 1,72,800           |                   |
| Add: Opening value of W-I-P           | 18,00,000          |                   |
| Less: Closing value of W-I-P          | (16,04,000)        |                   |
| <b>Cost of Production</b>             | <b>1,83,78,400</b> | <b>94.73</b>      |
| Add: Value of opening finished stock  | 9,60,000           |                   |
|                                       | (41,68,120)        |                   |

|                                                             |                    |               |
|-------------------------------------------------------------|--------------------|---------------|
| Less: Value of closing finished stock<br>(₹ 94.73 × 44,000) |                    |               |
| <b>Cost of Goods Sold</b>                                   | <b>1,51,70,280</b> | <b>94.81</b>  |
| Office and administration expenses paid                     | 26,52,000          | 16.58         |
| Travelling allowance paid to office staffs                  | 1,21,000           | 0.75          |
| Selling expenses                                            | 6,46,000           | 4.04          |
| <b>Cost of Sales</b>                                        | <b>1,85,89,280</b> | <b>116.18</b> |
| Add: Profit                                                 | 32,80,461          | 20.50         |
| Sales                                                       | <b>2,18,69,741</b> | <b>136.68</b> |

**Question – 5**

XYZ a manufacturing firm, has revealed following information for September, 2019:

|                   | 1 <sup>st</sup> September | 30 <sup>th</sup> September |
|-------------------|---------------------------|----------------------------|
|                   | ₹                         | ₹                          |
| Raw Materials     | 2,42,000                  | 2,92,000                   |
| Works-in-progress | 2,00,000                  | 5,00,000                   |

The firm incurred following expenses for a targeted production of 1,00,000 units during the month:

|                                                        | ₹           |
|--------------------------------------------------------|-------------|
| Consumable Stores and spares of factory                | 3,50,000    |
| Research and development cost for process improvements | 2,50,000    |
| Quality control cost                                   | 2,00,000    |
| Packing cost (secondary) per unit of goods sold        | 2           |
| Lease rent of production asset                         | 2,00,000    |
| Administrative Expenses (General)                      | 2,24,000    |
| Selling and distribution Expenses                      | 4,13,000    |
| Finished goods (opening)                               | Nil         |
| Finished goods (closing)                               | 5,000 units |

Defective output which is 4% of targeted production, realizes ₹ 61 per unit. Closing stock is valued at cost of production (excluding administrative expenses). Cost of goods sold, excluding administrative expenses amounts to ₹ 78,26,000. Direct employees' cost is ½ of cost of material consumed. Selling price of the output is ₹ 110 per unit.

You are required to:

- Calculate the value of material purchased
- Prepare cost sheet showing the profit earned by the firm

**Solution****Statement of cost and profit**

| Particulars                                                       | Amount             |
|-------------------------------------------------------------------|--------------------|
| Opening stock of material                                         | 2,42,000           |
| Add: Purchases (w.n. - 2)                                         | 52,50,000          |
| Less: Closing stock of material                                   | (2,92,000)         |
| <b>Direct material consumed</b>                                   | <b>52,00,000</b>   |
| Add: Direct Labour (w.n. - 2)                                     | 26,00,000          |
| Add: Direct Expenses                                              | -                  |
| <b>Prime Cost (bal. fig.)</b>                                     | <b>78,00,000</b>   |
| Add: Factory Overheads                                            |                    |
| Consumable stores and spares of factory                           | 3,50,000           |
| Lease rent of production asset                                    | 2,00,000           |
| <b>Gross Factory Cost (bal. fig.)</b>                             | <b>83,50,000</b>   |
| Add: Opening WIP                                                  | 2,00,000           |
| Less: Closing WIP                                                 | (5,00,000)         |
| <b>Net Factory Cost (bal. fig.)</b>                               | <b>80,50,000</b>   |
| Add: Quality Control Cost                                         | 2,00,000           |
| Add: Research and development Cost                                | 2,50,000           |
| Less: Sale of defective goods $(1,00,000 \times 4\% \times 61)$   | (2,44,000)         |
| <b>Cost of Production (bal. fig.)</b>                             | <b>82,56,000</b>   |
| Add: Opening stock of FG                                          | -                  |
| Less: Closing stock of FG (w.n. - 1)                              | (4,30,000)         |
| <b>Cost of goods sold</b>                                         | <b>78,26,000</b>   |
| Add: Administration overheads                                     | 2,24,000           |
| Add: Packaging cost (Secondary) $(2 \times 91,000)$               | 1,82,000           |
| Add: Selling & Distribution overheads                             | 4,13,000           |
| <b>Cost of Sales</b>                                              | <b>86,45,000</b>   |
| Add: Profit (bal. fig.)                                           | 13,65,000          |
| <b><math>[(1,00,000 - 4,000 - 5,000) \times 110]</math> Sales</b> | <b>1,00,10,000</b> |

**Working Notes:**

- (1) Since there is no opening stock, so the entire cost of goods sold is out of cost of production only.

$$\text{Thus, cost per unit of goods produced} = \frac{78,26,000}{1,00,000 - 5,000 - 4,000} = ₹ 86$$

$$\text{Therefore, closing stock of finished goods} = 86 \times 5,000 = ₹ 4,30,000$$

- (2) Let raw material purchase = y

$$\text{Thus, raw material consumed} = 2,42,000 + y - 2,92,000 = y - 50,000$$

$$\text{Direct wages} = \frac{1}{2} \times (y - 50,000) = 0.5y - 25,000$$

$$\text{Prime cost} = y - 50,000 + 0.5y - 25,000$$

$$78,00,000 = 1.5y - 75,000$$

$$1.5y = 78,75,000$$

$$y = 52,50,000$$

$$\text{Raw material purchased} = y = ₹ 52,50,000$$

### 9. Recovery Rate

It is the rate which is used to recover/absorb/charge overheads from the products being manufactured or services being provided.

Unless otherwise provided, following basis will be used for recovery of overheads:

|                                    |                  |
|------------------------------------|------------------|
| Factory overheads                  | Direct Labour    |
| Administration overheads           | NFC or work cost |
| Selling and distribution overheads | NFC or work cost |

#### Question – 6

The following figures are available from the books of SK Co. for the year 31<sup>st</sup> March:

|                                 | ₹      |                         | ₹      |
|---------------------------------|--------|-------------------------|--------|
| Materials:                      |        | Profit for the year     | 12,180 |
| Stock on 1 <sup>st</sup> April  | 2,000  | Selling overhead        | 10,500 |
| Stock on 31 <sup>st</sup> March | 4,000  | Factory overhead        | 9,000  |
| Purchases                       | 20,000 | Administration overhead | 8,400  |
| Wages                           | 15,000 |                         |        |

- (a) Prepare a cost sheet showing prime cost, work cost, cost of production, cost of sales and sales.
- (b) In April, the factory receives an order for a job which will require materials ₹ 2,400 and wages ₹ 1,500. Ascertain the sale price of the job if the factory intends to earn a profit 10% higher than the percentage of profit earned in year ending on 31<sup>st</sup> March. Assume that the factory overhead has gone up by 16(2/3)% and selling overhead has gone down by 20% after 31<sup>st</sup> March. Further assume that factory overhead is recovered as a percentage of the wages and administration and selling overhead as a percentage of works cost.

### Solution

#### Statement of Cost and Profit

| Particulars                     | Amount (₹) |
|---------------------------------|------------|
| Opening stock of material       | 2,000      |
| Add: Purchases                  | 20,000     |
| Less: Closing stock of material | (4,000)    |
| Direct material consumed        | 18,000     |

|                              |        |
|------------------------------|--------|
| Add: Direct wages            | 15,000 |
| Prime cost                   | 33,000 |
| Add: Factory overhead        | 9,000  |
| GFC/NFC/COP/COGS             | 42,000 |
| Add: Administration overhead | 8,400  |
| Add: Selling overhead        | 10,500 |
| Cost of Sales                | 60,900 |
| Add: Profit                  | 12,180 |
| Sales                        | 73,080 |

**Calculation of Recovery Rates**

Factory overheads as % of direct wages =  $\frac{(9,000 + 16.66666666\%)}{15,000} \times 100 = 70\%$  of direct wages

Administration overheads as % of NFC =  $\frac{8,400}{42,000} \times 100 = 20\%$  of NFC

Selling overheads as % of NFC =  $\frac{(10,500 - 20\%)}{42,000} \times 100 = 20\%$  of NFC

Profit as % of Cost of sales =  $\frac{(12,180 + 10\%)}{60,900} \times 100 = 22\%$  of Cost of sales

**Statement of calculation of selling price of Job**

| Particulars                                 | Amount (₹) |
|---------------------------------------------|------------|
| Direct Material                             | 2,400      |
| Direct wages                                | 1,500      |
| Prime Cost                                  | 3,900      |
| Add: Factory overheads (70% × 1,500)        | 1,050      |
| GFC/NFC/COP/COGS                            | 4,950      |
| Add: Administration overheads (20% × 4,950) | 990        |
| Add: Selling overheads (20% × 4,950)        | 990        |
| Cost of sales                               | 6,930      |
| Add: Profit (22% × 4,950)                   | 1,525      |
| Sales                                       | 8,455      |

**10. Change in Cost Effects**

Total Cost = No. Of units × Cost per unit

|                 | Total VC | Total FC |
|-----------------|----------|----------|
| Quantity Effect | Yes      | No       |
| Price Effect    | Yes      | Yes      |

Relation between:

- (A) Quantity and Variable cost : Direct relation  
 (B) Price and Variable cost : Direct relation



- (C) Price and Fixed cost : Direct relation  
 (D) Wages and Efficiency : Indirect relation

Unless otherwise provided, following points are to be assumed:

- (a) VC per unit will remain same  
 (b) Total FC will remain same  
 (c) All direct cost are considered to be variable in nature  
 (d) All overheads are considered to be fixed in nature

### Question – 7

A factory incurred the following expenditure during the year:

|                          | ₹                |
|--------------------------|------------------|
| Direct material consumed | 12,00,000        |
| Manufacturing wages      | 7,00,000         |
| Manufacturing overheads: |                  |
| Fixed                    | 3,60,000         |
| Variable                 | <u>2,50,000</u>  |
|                          | <u>6,10,000</u>  |
|                          | <u>25,10,000</u> |

In the next year, following changes are expected in production and cost of production.

- (a) Production will increase due to recruitment of 60% more workers in the factory.  
 (b) Overall efficiency will decline by 10% on account of recruitment of new workers.  
 (c) There will be an increase of 20% in fixed overhead and 60% in variable overhead.  
 (d) The cost of direct material will be decreased by 6%.  
 (e) The company desire to earn a profit of 10% on selling price.

Ascertain the cost of production and selling price.

### Solution

|                                                                |            |
|----------------------------------------------------------------|------------|
| Let existing production units                                  | 100        |
| Add: Increase due to recruitment of worker $(100 \times 60\%)$ | <u>60</u>  |
|                                                                | 160        |
| Less: Decline due to efficiency $(160 \times 10\%)$            | <u>16</u>  |
| New Production units                                           | <u>144</u> |

### Statement of cost and sale

| Particulars     | Working                                                               | Amount (₹) |
|-----------------|-----------------------------------------------------------------------|------------|
| Direct material | $\left(12,00,000 \times \frac{144}{100} \times \frac{94}{100}\right)$ | 16,24,320  |

|                                      |                                                                       |                  |
|--------------------------------------|-----------------------------------------------------------------------|------------------|
| Direct wages                         | $\left(7,00,000 \times \frac{144}{100} \times \frac{100}{90}\right)$  | 11,20,000        |
| <b>Prime Cost</b>                    |                                                                       | <b>27,44,320</b> |
| (+) Fixed manufacturing overheads    | $\left(3,60,000 \times \frac{120}{100}\right)$                        | 4,32,000         |
| (+) Variable manufacturing overheads | $\left(2,50,000 \times \frac{144}{100} \times \frac{160}{100}\right)$ | 5,76,000         |
| <b>Cost of Sales</b>                 |                                                                       | <b>37,52,320</b> |
| (+) Profit                           | (Bal. fig.)                                                           | 4,16,924         |
| <b>Sales</b>                         | $(37,52,320 \div 90\%)$                                               | <b>41,69,244</b> |

**Question – 8**

A factory's normal capacity is 1,20,000 units per annum. The estimated costs of production are as under:

- Direct material ₹ 3 per unit; direct labour ₹ 2 per unit (Subject to a minimum of ₹ 12,000 p.m.)
- Indirect expenses—Fixed ₹ 1,60,000 per annum: Variable ₹ 2 per unit; Semi-variable ₹ 60,000 upto 50% capacity and additional ₹ 20,000 for every 20% increase in capacity.
- Each unit of raw material yields scrap which is sold at the rate of 20 paise per unit.

The factory worked at 50% capacity for the first three months but it was expected that it would work @ 80% capacity for the remaining 9 months. During the first three months, the selling price per unit was ₹ 12. What should be the price in the remaining nine months to produce a total profit of ₹ 2,18,000?

**Solution****Statement of Cost**

| Particulars                  | First 3 months                                                                                         | Bal. 9 months                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Level of operation           | 50%                                                                                                    | 80%                                                                                                      |
| Units                        | $1,20,000 \times \frac{50}{100} \times \frac{3}{12} = 15,000$                                          | $1,20,000 \times \frac{80}{100} \times \frac{9}{12} = 72,000$                                            |
| Direct material @ ₹ 3 p.u.   | 45,000                                                                                                 | 2,16,000                                                                                                 |
| Direct wages                 | $\left\{ \begin{array}{l} 15,000 \times 2 \\ \text{or} \\ 12,000 \times 3 \end{array} \right\} 36,000$ | $\left\{ \begin{array}{l} 72,000 \times 2 \\ \text{or} \\ 12,000 \times 9 \end{array} \right\} 1,44,000$ |
| Fixed expenses               | $1,60,000 \times \frac{3}{12} = 40,000$                                                                | $1,60,000 \times \frac{9}{12} = 1,20,000$                                                                |
| Variable expenses @ ₹ 2 p.u. | 30,000                                                                                                 | 1,44,000                                                                                                 |
| Semi-variable expenses       | $60,000 \times \frac{3}{12} = 15,000$                                                                  | $(60,000 + 20,000 + 20,000) \times \frac{3}{12} = 75,000$                                                |
| (-) Scrap @ ₹ 0.20 p.u.      | (3,000)                                                                                                | (14,400)                                                                                                 |
| <b>Total Cost</b>            | <b>1,63,000</b>                                                                                        | <b>6,84,600</b>                                                                                          |

**Statement of Calculation of Selling Price for Remaining 9 Months**

|                                        |          |
|----------------------------------------|----------|
| Sales for first 3 months (15,000 × 12) | 1,80,000 |
| Less: Cost for first 3 months          | 1,63,000 |
| Profit for first 3 months              | 17,000   |
| Annual Target profit                   | 2,18,000 |
| Profit require from remaining 9 months | 2,01,000 |
| Add: Cost for remaining 9 months       | 6,84,600 |
| Sales for remaining 9 months           | 8,85,600 |
| Units for remaining 9 months           | 72,000   |
| Selling price for remaining 9 months   | 12.30    |

**Question – 9**

PNME Ltd. manufactures two types of masks – ‘disposal Masks’ and ‘Cloth Masks’. The cost data for the year ended 31<sup>st</sup> March, 2022 is as follows:

|                     | ₹         |
|---------------------|-----------|
| Direct materials    | 12,50,000 |
| Direct wages        | 7,00,000  |
| Production Overhead | 4,00,000  |
| Total               | 23,50,000 |

It is further ascertained that:

- Direct material cost per unit of cloth Mask was twice as much of direct material cost per unit of disposal Mask
- Direct wages per unit for Disposal Mask were 60% of those for Cloth Mask
- Production overhead per unit was at same rate for both the types of the masks
- Administration overhead was 50% of Production overhead for each type of mask
- Selling cost was ₹ 2 per cloth mask
- Selling price was ₹ 35 per unit of cloth mask
- No. of units of cloth masks sold – 45,000
- No. of units of Production of
  - o Cloth Masks : 50,000
  - o Disposal Masks : 1,50,000

You are required to prepare a cost sheet for cloth masks showing:

- (i) Cost per unit and total cost
- (ii) Profit per unit and total profit

**Solution****Preparation of Cost Sheet for Cloth Masks**

No. of units produced = 50,000 units

No. of units sold = 45,000 units

| Particulars                                           | Per unit (₹) | Total (₹) |
|-------------------------------------------------------|--------------|-----------|
| Direct materials (Working note (ii))                  | 10.00        | 5,00,000  |
| Direct wages (Working note (ii))                      | 5.00         | 2,50,000  |
| Prime cost                                            | 15.00        | 7,50,000  |
| Production overhead (Working note (iii))              | 2.00         | 1,00,000  |
| Factory Cost                                          | 17.00        | 8,50,000  |
| Administration Overhead* (50% of Production Overhead) | 1.00         | 50,000    |
| Cost of production                                    | 18.00        | 9,00,000  |
| Less: Closing stock (50,000 units – 45,000 units)     | –            | (90,000)  |
| Cost of goods sold i.e. 45,000 units                  | 18.00        | 8,10,00   |
| Selling cost                                          | 2.00         | 90,000    |
| Cost of sales/Total cost                              | 20.00        | 9,00,000  |
| <b>Profit</b>                                         | 15.00        | 6,75,000  |
| Sales value (₹ 35 × 45,000 units)                     | 35.00        | 15,75,000 |

**Working Notes:**

- (i) Direct material cost per unit of Disposable Mask = M  
 Direct material cost per unit of Cloth Mask = 2M  
 Total direct material cost =  $(2M \times 50,000) + (M \times 1,50,000)$   
 $12,50,000 = M \times 2,50,000$   
 $M = 12,50,000 \div 2,50,000 = ₹ 5$   
 Thus, direct material cost per unit of cloth mask =  $2 \times 5 = ₹ 10$
- (ii) Direct wages per unit of Cloth Mask = W  
 Direct wages per unit Disposable Mask = 0.6W  
 So,  $(W \times 50,000) + (0.6W \times 1,50,000) = ₹ 7,00,000$   
 $W = ₹ 5$  per unit  
 Therefore, Direct material Cost per unit of Cloth Mask = ₹ 5

- (iii) Production overhead per unit =  $\frac{4,00,000}{(50,000+1,50,000)} = ₹ 5$

Production overhead for Cloth Mask = ₹ 2 × 50,000 units = ₹ 1,00,000

\*Administration overhead is related to production overhead in the question and hence to be considered in cost of production only.

**Question – 10**

MNP Limited have the capacity to produce 84,000 units of a product very month. Its prime cost per unit at various levels of production is as follows:

| Level | Prime Cost per unit (₹) |
|-------|-------------------------|
| 10%   | 50                      |
| 20%   | 48                      |
| 30%   | 46                      |
| 40%   | 44                      |
| 50%   | 42                      |
| 60%   | 40                      |

|      |    |
|------|----|
| 70%  | 38 |
| 80%  | 36 |
| 90%  | 34 |
| 100% | 32 |

Its prime cost consists of raw material consumed, direct wages and direct expenses in the ratio of 3:2:1. In the month of January 2024, the company worked at 40% capacity and raw material purchased amounting to ₹8,40,000. In the month of February 2024, the company worked at 100% capacity and raw material purchased for ₹16,46,400.

It is the policy of the company to maintain opening stock of raw material equal to 1/3 of closing stock of raw material. Factory overheads are recovered at 60% of direct wages cost. Fixed administration expenses (as part of production cost) and fixed selling and distribution expenses are ₹2,01,600 and ₹1,68,000 per month respectively. During the month of January 2024 company sold 33,600 units @ ₹68.8 per unit. The variable distribution cost amounts to ₹1.5 per unit sold.

The management of the company chalks out a plan for the month of February 2024 to sell its whole output @ ₹61 per unit by incurring following further expenditure:

- Company sponsors a television programme on every Sunday at a cost of ₹26,250 per week. There are 4 Sundays in February 2024.
- Hi-tea programme every month for its potential customers at a cost of ₹1,05,000.
- Special gift item costing ₹105 on sale of a dozen units.
- Lucky draws scheme is introduced every month by giving the first prize of ₹1,00,000; second prize of ₹80,000; third prize of ₹40,000 and four consolation prizes of ₹8,000 each.

**Note:** (In the month of February 2024, there is a significant saving in material cost per unit due to entry of new suppliers in the market and saving in per unit cost of Direct wages and Direct expenses due to introduction of new policy by the management.)

Prepare a cost sheet for the month of January 2024 and February 2024 showing prime cost (with different elements of prime cost), factory cost, cost of production, total cost and profit earned.

### Solution

#### Cost Sheet

| Particulars                             | January 2024     | February 2024    |
|-----------------------------------------|------------------|------------------|
|                                         | 33,600 Units     | 84,000 Units     |
| Opening Stock of Raw Material           | 50,400           | 1,51,200         |
| Add: Purchases                          | 8,40,000         | 16,46,400        |
| Less: Closing stock of Raw Material     | (1,51,200)       | (4,53,600)       |
| <b>Direct materials consumed:</b>       | <b>7,39,200</b>  | <b>13,44,000</b> |
| Direct Wages                            | 4,92,800         | 8,96,000         |
| Direct expenses                         | 2,46,400         | 4,48,000         |
| <b>Prime Cost</b>                       | <b>14,78,400</b> | <b>26,88,000</b> |
| Factory overheads (60% of direct wages) | 2,95,680         | 5,37,600         |

|                                                 |                  |                  |
|-------------------------------------------------|------------------|------------------|
| <b>Factory / Works Cost</b>                     | <b>17,74,080</b> | <b>32,25,600</b> |
| Add: Administration overhead (Production)       | 2,01,600         | 2,01,600         |
| <b>Cost of Production / Cost of goods sold</b>  | <b>19,75,680</b> | <b>34,27,200</b> |
| Add: Fixed selling and distribution Overhead    | 1,68,000         | 1,68,000         |
| Variable distribution overheads (₹1.5 per unit) | 50,400           | 1,26,000         |
| Sponsorship cost                                | —                | 1,05,000         |
| Hi tea programme                                | —                | 1,05,000         |
| Special gifts (84,000 × 1/12 × 105)             | —                | 7,35,000         |
| Lucky draw prize*                               | —                | 2,52,000         |
| <b>Cost of sales / Total Cost</b>               | <b>21,94,080</b> | <b>49,18,200</b> |
| Profit (Balancing figure)                       | 1,17,600         | 2,05,800         |
| <b>Sales revenue</b>                            | <b>23,11,680</b> | <b>51,24,000</b> |

**\*Lucky draw prize:**

|                                 | <b>Amount (₹)</b> |
|---------------------------------|-------------------|
| 1 <sup>st</sup> Prize           | 1,00,000          |
| 2 <sup>nd</sup> Prize           | 80,000            |
| 3 <sup>rd</sup> Prize           | 40,000            |
| Consolation Prizes (4 × ₹8,000) | 32,000            |
| <b>Total</b>                    | <b>2,52,000</b>   |

**Working note:****Calculation of opening and costing stock of Raw Material****January**

Units Manufactured = 84,000 × 40% = 33,600 units  
 Prime Cost = 33,600 × 44 = ₹14,78,400  
 Raw Material consumed = ₹14,78,400 × 3/6 = ₹7,39,200  
 Raw Material purchase (given) = ₹8,40,000

Let closing stock of Raw Material be x

Opening stock of Raw Material be 1/3x

Opening Stock + Purchase – closing stock = Raw Material consumed

$1/3x + ₹8,40,000 - x = ₹7,39,200$

$1/3x - x = ₹7,39,200 - ₹8,40,000$

$2/3x = ₹1,00,800$

$x = ₹1,51,200$  (closing stock)

Opening stock = ₹1,51,200 × 1/3 = ₹50,400

**February**

Prime Cost = 84,000 × 32 = ₹26,88,000

Raw Material consumed = ₹26,88,000 × 3/6 = ₹13,44,000

Raw Material purchased (given) = ₹16,46,400

Opening Stock + Purchase – closing stock = Raw Material consumed

$₹1,51,200 + ₹16,46,400 - \text{closing stock} = ₹13,44,000$

Closing stock = ₹4,53,600